

Local Food Wastage Management System

1. Problem Statement

Food wastage is a major challenge in society, where restaurants, supermarkets, and individuals often dispose of surplus food while many people face food insecurity. The Local Food Wastage Management System aims to connect food providers with receivers through a centralized platform, reducing food waste and improving food distribution.

2. Business Use Cases

- Connect surplus food providers with NGOs and individuals in need.
- Reduce food wastage through efficient redistribution.
- Improve accessibility to available food resources.
- Analyze food donation and claim trends for better decision-making.
- Support sustainable food management practices.

3. Dataset Description

The project uses four datasets:

Providers Dataset

Contains details of food providers such as restaurants, supermarkets, and grocery stores.

Receivers Dataset

Contains details of NGOs, community centers, and individuals receiving food.

Food Listings Dataset

Contains information about available food items, quantity, location, food type, and meal type.

Claims Dataset

Contains details of food claims made by receivers and claim status.

4. Database Design

The SQL database consists of four tables:

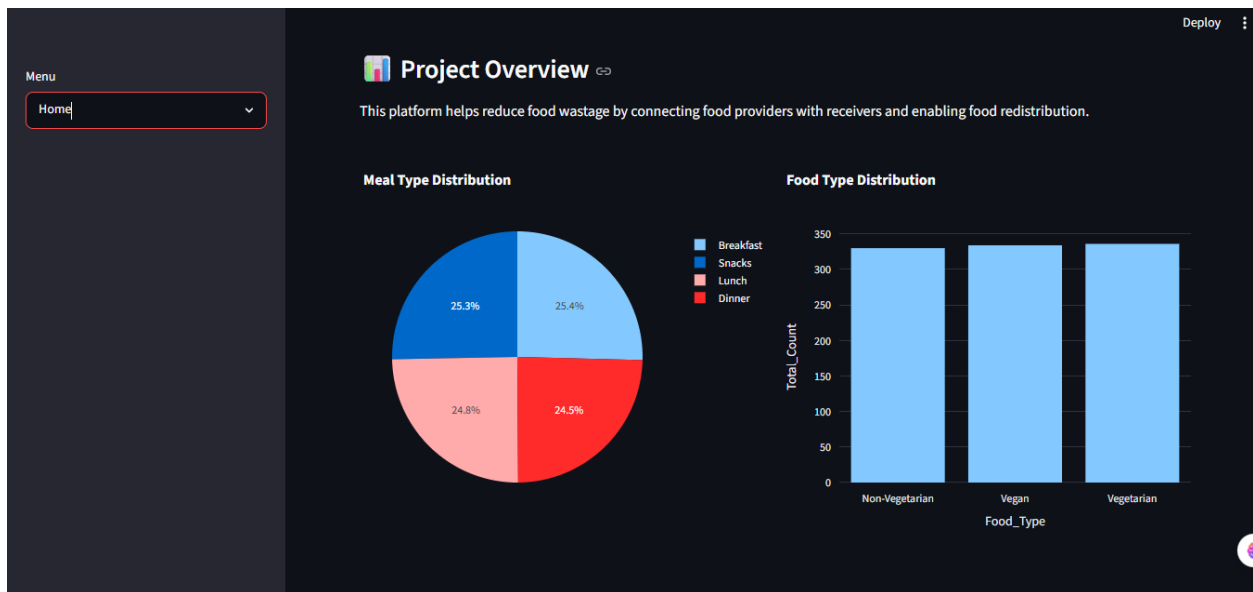
- Providers
- Receivers
- Food_Listings
- Claims

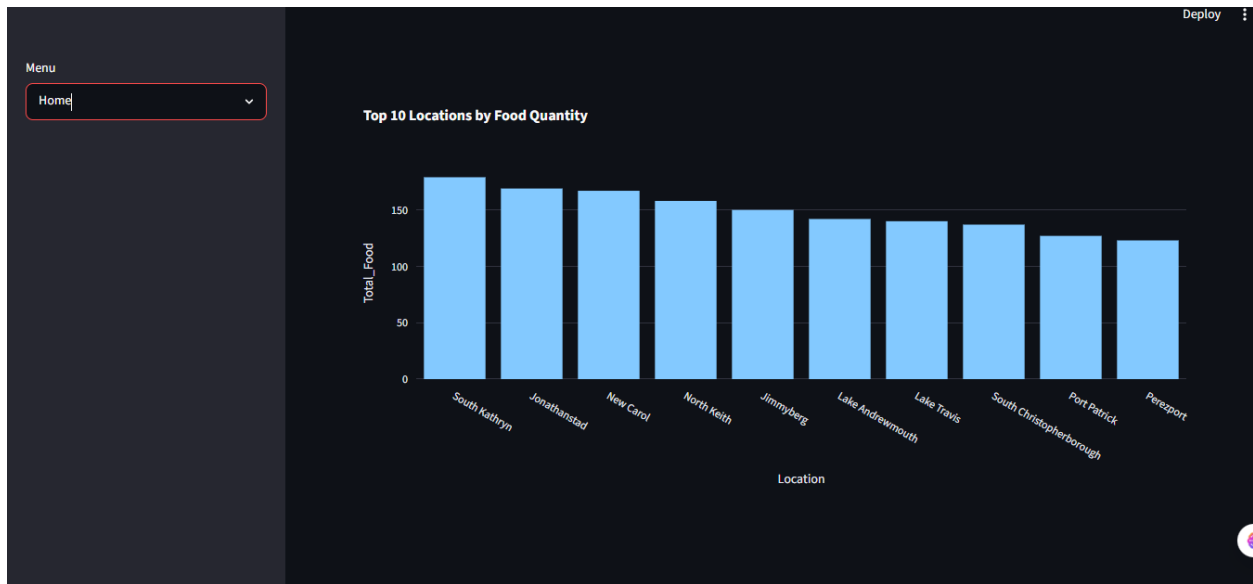
Relationships:

- Providers provide food listings.
- Receivers claim food items.
- Claims table connects food listings and receivers.

5. Exploratory Data Analysis (EDA)

The dataset was analyzed to understand food donation patterns and food distribution trends.





Key Findings:

- Certain meal types contribute more food donations.
- Some provider categories donate significantly more food.
- A few locations contribute the highest food quantity.

6. SQL Query Outputs

Fifteen SQL queries were executed to analyze:

1. Total Food Available
2. Food Listings by Location
3. Food Type Distribution
4. Provider Type Contribution
5. Providers by City
6. Receivers by City
7. Providers Contact List
8. Top Receivers by Claims
9. Most Claimed Food Items
10. Providers with Successful Claims
11. Average Food Quantity

12. Meal Type Distribution
13. Top Providers by Food Donation
14. Top Locations by Food Quantity
15. Provider Type Performance

7. Streamlit Dashboard

The Streamlit application provides:

- Home Dashboard
- Providers Data
- Receivers Data
- Food Listings
- Filters
- SQL Dashboard
- Add Food
- Update Food
- Delete Food



Menu

Providers

Deploy

Providers Data

	Provider_ID	Name	Type	Address	City
0	1	Gonzales-Cochran	Supermarket	74347 Christopher Extensions Andreamouth, OK 91839	New Jessica
1	2	Nielsen, Johnson and Fuller	Grocery Store	91228 Hanson Stream Welchtown, OR 27136	East Sheena
2	3	Miller-Black	Supermarket	561 Martinez Point Suite 507 Guzmanchester, WA 94320	Lake Jesusview
3	4	Clark, Prince and Williams	Grocery Store	467 Bell Trail Suite 409 Port Jesus, IA 61188	Mendezmouth
4	5	Coleman-Farley	Grocery Store	078 Matthew Creek Apt. 319 Saraborough, MA 53978	Valentineside
5	6	Lawson-Walters	Grocery Store	1889 Barnes Gateway Adamview, ID 87971	Shannonside
6	7	Ortiz-Lee	Restaurant	1842 Villarreal Shores Williamfort, CT 44529	Lake Christopherburgh
7	8	Curtis-Lewis	Supermarket	4770 Miller Light Suite 260 New Charlesville, AR 97075	Washingtonville
8	9	Nguyen Inc	Supermarket	169 Scott Keys Suite 000 East Michelle, HI 57025	Tinamouth
9	10	Hall, Nguyen and Martinez	Catering Servic	71458 Mark Field Apt. 252 North Alexland, TX 02693	West Cherylfort

Menu

Food Listings

Deploy

Food Listings

	Food_ID	Food_Name	Quantity	Expiry_Date	Provider_ID	Provider_Type	Location	Food_Type	Meal_Ty
0	1	Bread	43	3/17/2025	110	Grocery Store	South Kellyville	Non-Vegetarian	Breakfas
1	2	Soup	22	3/24/2025	791	Grocery Store	West James	Non-Vegetarian	Dinner
2	3	Fruits	46	3/28/2025	478	Catering Service	Lake Regina	Vegan	Breakfas
3	4	Fruits	15	3/16/2025	930	Restaurant	Kellytown	Vegan	Lunch
4	5	Soup	14	3/19/2025	279	Restaurant	Garciaport	Vegan	Dinner
5	6	Vegetables	4	3/24/2025	378	Supermarket	South Johnshire	Vegan	Snacks
6	7	Dairy	29	3/25/2025	268	Grocery Store	Nolanmouth	Vegan	Dinner
7	8	Rice	49	3/28/2025	343	Catering Service	South Christopherborough	Vegetarian	Lunch
8	9	Bread	46	3/25/2025	47	Supermarket	East Andrewland	Non-Vegetarian	Snacks
9	10	Rice	41	3/20/2025	528	Supermarket	Markport	Vegan	Dinner

Menu

Receivers

Deploy

Receivers Data

	Receiver_ID	Name	Type	City	Contact
0	1	Donald Gomez	Shelter	Port Carlburgh	(955)922-5295
1	2	Laurie Ramos	Individual	Lewisburgh	761.042.1570
2	3	Ashley McKee	NGO	South Randalltown	691-023-0094x856
3	4	Erika Rose	NGO	South Shaneville	8296491111
4	5	John Romero	Individual	Bakerport	067.491.0154
5	6	Mandy Sutton PhD	Shelter	East Sharimouth	682-777-5357
6	7	Kenneth Baker	NGO	South Edwinborough	126-645-6386x4071
7	8	James Perez	Charity	Benjaminburgh	-8421
8	9	Emily Turner	Shelter	West Robert	001-631-022-6157x520
9	10	Mary Salazar	NGO	Wrightland	001-187-333-7518x11395

Menu

Filters

Deploy

Filters

Select City

Adambury

Select Provider Type

Catering Service

Select Food Type

Non-Vegetarian

Select Meal Type

Lunch

	Food_ID	Food_Name	Quantity	Expiry_Date	Provider_ID	Provider_Type	Location	Food_Type	Meal_Ty
426	427	Soup	34	3/18/2025	338	Catering Service	Adambury	Non-Vegetarian	Lunch

Menu

SQL Dashboard

Deploy

SQL Dashboard

1. Total Food Available

	Total_Food_Available
0	25794

2. Food Listings by Location

	Location	Total_Listings
0	New Carol	6
1	South Kathryn	6
2	Jimmyberg	5
3	Perezport	5
4	East Angela	5
5	South Christopherborough	4
6	North Keith	4
7	East Quabok	4

Menu

Add Food

Deploy

Add Food Listing

Food Name

Pani Puri

Quantity

1

Expiry Date

2026/06/19

Provider ID

1

Provider Type

Restaurant

Location

Menu

Update Food

Deploy

Update Food Quantity

Enter Food ID

3

New Quantity

2

Update Food

Menu

Delete Food

Deploy

Delete Food Listing

Enter Food ID To Delete

6

Delete Food

8. Insights

- Food donations are concentrated in specific cities.
- Certain provider types contribute larger quantities of food.
- Some food categories are more frequently available.
- Receiver claim activity varies significantly across locations.
- Effective redistribution can substantially reduce food wastage.

9. Recommendations

- Encourage more restaurants and supermarkets to participate.
- Prioritize food distribution in high-demand locations.
- Implement automated notifications for newly available food.
- Improve tracking of food claims and completion rates.
- Expand the platform to additional cities and organizations.

10. Conclusion

The Local Food Wastage Management System successfully connects food providers and receivers through a centralized platform. The project demonstrates how SQL, Python, and Streamlit can be used to reduce food wastage, improve food accessibility, and generate valuable insights through data analysis.

11. Streamlit Application Link

<https://local-food-wastage-management-system-mdrddwropfy3qxh3kpdmgv.streamlit.app/>